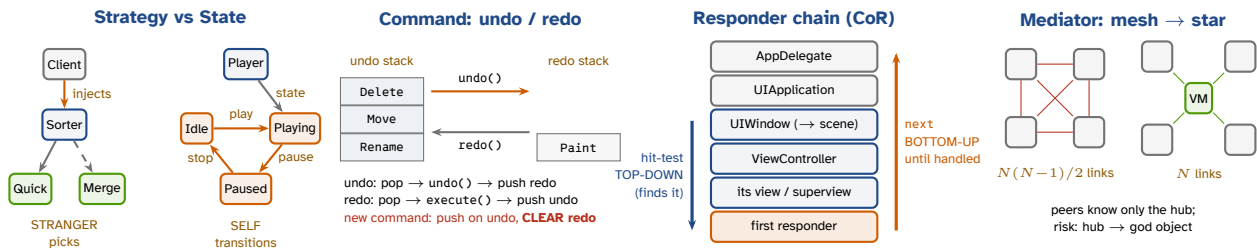


In one line: Behavioural patterns decide *who talks to whom and who decides*: **Strategy** (client picks the algorithm) · **State** (object switches itself) · **Observer** (one-to-many notify) · **Command** (request as object, execute+undo) · **Template Method** (base skeleton, subclass hooks) · **Chain of Responsibility** (pass until handled) · **Mediator** (hub instead of mesh) · **Iterator** (next() until nil).



Strategy vs State — “stranger picks vs self transitions”

Strategy: a family of interchangeable algorithms behind one protocol; the **CLIENT** chooses and injects one; strategies don't know each other. **iOS:** `sorted(by:)` closure, `JSONDecoder.keyDecodingStrategy`.

State: behaviour changes with internal state, and the **states themselves trigger the transition** (`context.state = Paused()`); each knows its successors. **iOS:** `player / download / connection lifecycle`; in Swift often an enum + switch. Same class diagram — differ in *who changes it*.

Observer Subject notifies **many** subscribers without knowing their types (delegation is the 1:1, answer-back sibling).

iOS: `NotificationCenter` · `KVO` (`NSObject + @objc dynamic`) · `Combine` (`@Published` fires on `willSet`) · `@Observable` (iOS 17, per-property tracking).

The problem it CAUSES — leaks / retain cycles: publisher keeps your closure, closure captures self, self keeps the subscription → cycle; forgotten observers keep firing. Fix: `sink { [weak self] in ... }.store(in: &bag)`; `removeObserver(token)` for block-based `NotificationCenter`; `NSKeyValueObservation` stops when released.

Command A request wrapped as an **object** with `execute()` **AND** `undo()` — so it can be queued, logged, retried, sent later, or **undone**. **Undo/redo = two stacks** (diagram above).

iOS: `UndoManager` (`registerUndo(withTarget:handler:)`), `UIAction`, `Operation + OperationQueue`, `target-action`.

Template Method Base class owns the algorithm's **skeleton** and calls **overridable hooks** that subclasses fill in. **Hollywood Principle:** “don't call us, we'll call you”.

iOS: `UIViewController lifecycle` — `viewDidLoad`, `viewWillAppear`, `viewDidLayoutSubviews` (call super); `UIView.draw(_:)`, `layoutSubviews`. **vs Strategy = inheritance vs composition:** TM varies steps via subclassing (fixed at compile time); Strategy swaps the *whole* algorithm at runtime.

Chain of Responsibility Pass a request along handlers **until one handles it**; sender doesn't know who; may fall off the end.

iOS: responder chain: hit-testing (`hitTest(_:with:)`) runs **TOP-DOWN** from the window to the deepest view; the unhandled event then goes **BOTTOM-UP** via `next`: `view` → `its VC` → `superview` ... → `window` → `UIApplication` → `app delegate`. `sendAction(_:to: nil, ...)` walks it too.

Trap: Decorator: every wrapper runs; CoR: *one* decides.

Mediator Colleagues talk only to a **hub**: $N(N-1)/2$ peer links collapse to N . **iOS: ViewModel, Coordinator** (VCs never push each other), `UINavigationController`. **vs Facade:** Facade simplifies a subsystem, one-way, subsystem unaware; Mediator coordinates peers *both ways*. Risk: god object.

Iterator Walk a collection without exposing its storage.

Sequence (`makeIterator()`) + **IteratorProtocol** (`mutating func next() → Element?, nil = end`). for `x in s` ≡ `var it = s.makeIterator(); while let x = it.next(). for-in` needs only Sequence; Collection adds multi-pass + indices. **AsyncSequence:** `next()` async throws, for `try await` (`URL.lines`, `AsyncStream`).

Remember Strategy: stranger picks · State: self transitions · Template: don't call us · CoR: hit-test down, next up · Mediator: $N^2 \rightarrow N$ · Command: execute + undo, two stacks · Iterator: nil ends it.

```
protocol Command { func execute(); func undo() }
final class History {
    // the invoker
    private var undoStack: [Command] = [], redoStack: [Command] = []
    func run(_ c: Command) {
        c.execute(); undoStack.append(c); redoStack.removeAll()
    }
    func undo() { guard let c = undoStack.popLast() else { return }
        c.undo(); redoStack.append(c) }
    func redo() { guard let c = redoStack.popLast() else { return }
        c.execute(); undoStack.append(c) }
}
// Observer without the leak: weak capture + stored token
vm.$title.sink { [weak self] in self?.label.text = $0 }
.store(in: &bag) // drop it = never fires
```

Traps

- ! Strategy ≠ State: *who* changes it — client vs the state itself.
- ! Observer's own failure mode is the **leak**: strong self in a stored closure.
- ! Command without `undo()` is just a callback.
- ! Responder chain goes **UP**; hit-testing goes **DOWN**.
- ! Mediator ≠ Facade: two-way peers vs one-way simplification.